

[TARGET DRAFT — NOT FOR SUBMISSION]

Marginal Matching Does Not License Factorized Sampling: Auditing Conditional Style Leakage in Factorized Generative Models

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Target draft — projected values, not results. This is a *planning* draft showing what the paper is aimed at, not what it currently reports. Values in blue with a dagger[†] are **projected best-case outcomes of experiments that have not been run**; every other value is measured. The submission file is `main_v6.tex`, which contains measured values only. Sections marked [†] in their heading do not yet exist as experiments. Before any submission, every [†] value must be replaced by a measurement and the `\pj` macro deleted.

Abstract

Factorized generative models regularize a style latent z_s toward a fixed prior using a marginal statistic, $q(z_s) \approx \mathcal{N}(0, I)$, and treat this as a certificate that z_s is class-invariant. We show the certificate is invalid: matching the marginal places no constraint on the class conditionals $q(z_s | y=k)$, so z_s can be exactly Gaussian in aggregate while remaining maximally informative about the class label y . We call the resulting failure *conditional style leakage* and give a diagnostic for it based on inter-class style separation and linear probing. Across four unsupervised baselines (VAE, WAE-MMD, β -TCVAE, FactorVAE) and a case-study model, F-CS-WAE, every model with near-zero global MMD nonetheless allows a linear probe to recover y from z_s with 74–100% accuracy (chance 10%); on MNIST, F-CS-WAE reaches 99.15% clustering accuracy while class-conditional generation succeeds only 15% of the time under the sampling procedure that marginal matching is supposed to license. No single one of six one-at-a-time perturbations (latent capacity, training curriculum, prior geometry, supervision) eliminates the leakage on either MNIST or CIFAR-10, so it is not attributable to any one of these design choices alone – though on CIFAR-10 its magnitude is highly sensitive to capacity, which a MNIST-only check would have missed. **Comparing four remedies that attack the leakage through different mechanisms isolates the decomposition’s central prediction: gradient reversal cuts linear-probe recovery to 21%, less than half what per-class style MMD achieves, yet naive factorized sampling still fails one time in five – eliminating conditional leakage is necessary but not sufficient, because within-class semantic–style dependence survives every one of them. A remedy that targets that fourth term instead lifts naive sampling to 91% while leaving the**

probe higher.[†] We test two sampling-side remedies. A supervised per-class style MMD substantially reduces, but does not eliminate, the leakage, at a clustering cost that is dataset-dependent rather than a fixed price of the fix. A post-hoc class-conditional style prior fully repairs MNIST generation without retraining, but only partially repairs CIFAR-10, so a remedy validated on one dataset does not automatically generalize to another. The broader point is methodological: no divergence computed on $q(z_s)$ alone can certify $z_s \perp y$, and factorized generative models that report only marginal statistics have not actually verified the property they claim.

1 Introduction

Matching the aggregate style posterior to a Gaussian prior is a statement about a marginal distribution; class-invariant style is a statement about a family of conditional distributions. The two are not the same claim, and satisfying the first does not satisfy the second: $q(z_s)$ can be exactly $\mathcal{N}(0, I)$ while each $q(z_s | y=k)$ occupies a different region of latent space, so long as those regions average out to the right aggregate shape. A regularizer that only ever looks at $q(z_s)$ has no way to see this, and therefore no way to prevent it.

Where this bites. Consider a toy version of the standard factorized setup: a semantic variable z_c carries class identity, a style variable z_s is meant to carry everything else, and z_s is trained to satisfy $q(z_s) \approx \mathcal{N}(0, I)$. During training, the decoder only ever sees (z_c, z_s) pairs drawn from the same image, so if $q(z_s | y=k)$ happens to differ across classes—which nothing in the training objective prevents—the decoder is free to learn class-specific interactions between z_c and z_s . At generation time, a class prior supplies z_c for class k while z_s is drawn from the *marginal*, $\mathcal{N}(0, I)$, which averages over all classes. The resulting pair (z_c, z_s) is one the decoder never saw during training, and its output is unconstrained by anything the model learned. We call this failure *conditional style leakage*.

Why the usual metrics do not catch it. The failure is silent precisely because it is invisible to the metrics a practitioner

would normally check. Reconstruction quality is measured on the posterior manifold – z_s inferred from a real image – where the class-specific interaction the decoder learned is exactly the one being exercised, so reconstruction looks fine. Global MMD is a marginal statistic by definition and, as above, is compatible with arbitrary conditional structure. Clustering accuracy depends only on the semantic variable z_c and says nothing about z_s at all. A model can therefore score well on every diagnostic a typical evaluation protocol reports and still fail completely at class-conditional generation, which is the one capability the factorization exists to provide.

A diagnostic, not just an example. Because the failure does not show up in standard metrics, detecting it requires measurements taken directly on the conditional structure of z_s : we use inter-class style separation Δ_{inter} and linear probing accuracy of y from z_s (Section 3.5, which also defines a complementary kernel-based diagnostic, $\text{HSIC}(z_s, y)$), alongside the marginal MMD that current practice already reports. A model with near-zero MMD but large Δ_{inter} or elevated linear-probe accuracy is leaking class information through a channel that was supposed to be class-invariant.

What we find. Applying this diagnostic across five model families on MNIST, every model with near-zero global MMD still allows a linear probe to recover the class label from z_s with 74–100% accuracy against a 10% chance rate (Section 6.1); this includes β -TCVAE and FactorVAE, which are explicitly designed to penalize latent dependence. As a concentrated case study, we instantiate F-CS-WAE (Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoder), a model that separates a hyperspherical semantic variable from a Euclidean style variable; without any conditional regularization it reaches 99.15% clustering accuracy on MNIST while class-conditional generation succeeds only 15% of the time. We then test two remedies – a supervised per-class style MMD applied at training time, and a post-hoc class-conditional style prior applied at sampling time – and report where each helps, where it does not fully fix the problem, and where a fix validated on MNIST fails to transfer to CIFAR-10 (Section 6.7).

Scope. We use F-CS-WAE to exhibit and repair the failure, not to argue it is a better model than the baselines it is compared against: the cross-model comparison in Section 6.1 does not control for encoder capacity or training budget, the CIFAR-10 baseline comparison in Section 5 is single-seed, and the proposed per-class style MMD is a supervised, label-guided regularizer, not an unsupervised disentanglement method. These scope limits are stated where they arise and collected in Section 7.1.

Contributions. Our contributions are fourfold. **(1) A decomposition of factorized-sampling validity.** We show the mismatch a factorized sampler incurs decomposes exactly into four conceptually distinct, non-negative terms (Theorem 1): global style-prior mismatch, conditional style leakage, semantic-prior mismatch, and within-class semantic-style dependence. Two corollaries make the consequence precise: valid factorized sampling requires all four terms to vanish (so class-invariant style is necessary but not sufficient, Corollary 1), and the latent-space mismatch upper-bounds the resulting image-space mismatch but not conversely (Corollary 2), which is what makes a decoder-level intervention necessary rather than redundant. **(2) A conditional audit protocol.** We build a lightweight suite of finite-sample proxies for the four terms and apply it across five model families, showing the conditional leakage the decomposition predicts is real even when – for three of the five – it is invisible to a naive 2D projection of the latent (Section 6.1, Appendix D). **(3) A decoder-sensitivity and robustness study.** A controlled latent-swap intervention shows the decoder is measurably more sensitive to the leaked style code than to the semantic code, and that this sensitivity shifts once the remedy is applied (Section 6.4); a six-way, one-at-a-time perturbation of capacity, curriculum, prior geometry, and supervision on both datasets shows no single design choice eliminates the leakage, while revealing that its magnitude is strongly capacity-sensitive on CIFAR-10 in a way a MNIST-only check would miss (Section 6.9). **(4) An evaluation of representation- and sampling-level repairs, with a negative result.** We study per-class style MMD – itself a direct, conditional-matching instantiation of the variational-fair-autoencoder idea [19] applied to a style code, not a new mechanism – and a post-hoc class-conditional prior, and report where each helps, where it does not, and a negative transfer result: the sampling recipe that fully repairs MNIST generation only partially repairs CIFAR-10 (Sections 4, 6.8, 6.7). **Comparing four mechanistically different term-2 remedies at matched budget (Section 6.10) supplies the decomposition’s missing empirical test: they span 21–46% linear-probe recovery yet all land within 0.735–0.79 naive self-accuracy and leave term 4 untouched, dissociating conditional leakage from generation quality exactly as Corollary 1 predicts; a remedy targeting term 4 instead reaches 0.91.[†]**

2 Related Work

We organize prior work around a single question: does the method’s training or evaluation protocol certify conditional style invariance, $z_s \perp y$, or only a marginal or joint statistic that is compatible with $z_s \not\perp y$?

Variational and Wasserstein auto-encoders. VAEs [13] impose a per-example KL penalty toward a fixed Gaussian

prior; WAEs [23] relax this to a divergence between the *aggregated* posterior and the prior, computed with kernel two-sample tests [5] or projected distances [15]. The VAE penalty is not purely an aggregate regularizer in disguise: averaged over the data distribution it decomposes exactly as $\mathbb{E}_{q(x)}[\text{KL}(q(z|x) \| p(z))] = I_q(X; Z) + \text{KL}(q(z) \| p(z))$ [9], an information-capacity term plus an aggregate-prior-mismatch term structurally identical to what WAE-MMD targets directly. Neither term, however, certifies $I(z_s; y) = 0$: the mutual-information term bounds how much Z encodes about X in general, not specifically about the label y , and the aggregate term is exactly the marginal statistic Proposition 1 shows is insufficient. WAE-MMD matches the aggregate posterior only; VAE’s per-example KL is an information-bottleneck term plus that same aggregate match; β -TCVAE/FactorVAE below penalize dependence *among latent dimensions*, a third, still-different target. None of these three families evaluates $q(z_s | y)$, so none certifies conditional invariance even when its own target is well matched.

Disentanglement and total correlation. β -VAE [8] increases the KL weight to encourage disentangled factors; β -TCVAE [1] and FactorVAE [12] isolate and penalize the total correlation between latent dimensions directly. These methods get closer to our concern, since total correlation is a joint-distribution statistic rather than a purely marginal one, but the joint they penalize is over *latent dimensions*, not over the latent and the label. Table 3 shows both still leak: β -TCVAE and FactorVAE have among the lowest global MMD in our comparison, yet a linear probe recovers the class label from their style latent at 73.9% and 75.9% accuracy respectively, far above the 10% chance rate.

Conditional and semi-supervised generative models. Semi-supervised VAEs [14] and CVAE [22] condition generation on the label directly, which sidesteps needing z_c to encode class identity on its own, but neither checks whether the *residual* latent still carries label information after conditioning. StyleGAN [11] factorizes coarse structure from fine style via a mapping network in a class-unconditional regime, so the question does not arise there in the same form. Our setting requires both a class-conditional generator and a label-free clustering signal from the same model, which is what makes residual leakage in z_s consequential rather than incidental.

Deep generative clustering. DEC [25], IDEC [6], and VaDE [10] evaluate almost exclusively via ACC/NMI/ARI on the semantic or cluster-assignment variable. This protocol cannot see conditional style leakage by construction, since it never inspects z_s : our MNIST case study makes this concrete, reaching 99.15% clustering accuracy (Section 6.1) on exactly the checkpoint whose class-conditional generation succeeds only 15% of the time (Section 6.3).

Invariant and nuisance-free representations. Outside the generative-modeling literature, several methods target exactly the dependency this paper studies, $z_s \not\perp y$, through mechanisms other than an aggregate divergence. The variational fair autoencoder [19] matches a nuisance-conditional posterior toward a common target per nuisance value – the per-class style MMD we use in Section 4.2 is a direct instantiation of this idea applied to a style code, not a new mechanism. Gradient-reversal domain adaptation [3] and Fader Networks [17] instead remove label information from a representation adversarially, in image or feature space; Moyer et al. [20] achieve a similar effect without an adversary via a different information-theoretic penalty. None of these methods evaluates class-conditional generation under a factorized sampler of the kind Section 3 analyzes, so none of them establishes whether removing $z_s \not\perp y$ alone is sufficient to make independent sampling reliable – the question Section 3.3 shows it is not, since it leaves within-class z_c - z_s dependence untouched.

Hyperspherical latent variables. S-VAE [2] replaces the Gaussian posterior with a von Mises-Fisher distribution; Spherical Cauchy VAEs [21] give a heavier-tailed alternative with a Möbius-transform reparameterization. We use the latter for the semantic variable in our case-study model (Section 4.3). This choice of geometry is an implementation detail of the case study, not a claim about the leakage problem, which is defined independently of how z_c is parameterized.

3 The Conditional Style Leakage Problem

3.1 Setting and Notation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled dataset with K classes. A factorized model encodes each input into a semantic variable z_c and a style variable z_s . The style variable is regularized by

$$\mathcal{L}_{\text{style}} = D(q(z_s), \mathcal{N}(0, I)), \quad (1)$$

where D is any aggregate divergence and $q(z_s) = \mathbb{E}_y[q(z_s | y)]$. Class-conditional generation samples $z_s \sim \mathcal{N}(0, I)$ independently of z_c .

3.2 Formal Gap

Proposition 1 (Marginal matching does not certify conditional independence). *Let $z_s \sim \mathcal{N}(0, I)$ on $\mathbb{R}^{d_s}$, and let $\{A_k\}_{k=1}^K$ be any measurable partition of $\mathbb{R}^{d_s}$ with $P(z_s \in A_k) = 1/K$ for each k (such a partition always exists, e.g. via level sets of a continuous statistic of z_s). Define the label y as the deterministic function $y = k \iff z_s \in A_k$. Then:*

- (i) *the marginal is unchanged, $q(z_s) = \mathcal{N}(0, I)$ exactly, so $D(q(z_s), \mathcal{N}(0, I)) = 0$ for every divergence D ; and*

(ii) $I(z_s; y) = H(y) = \log K$, the maximum mutual information achievable between z_s and a K -ary label.

Proof. (i) y is defined post hoc as a measurable function of z_s ; the distribution of z_s itself is never altered, so $q(z_s) = \mathcal{N}(0, I)$ holds exactly and every divergence to $\mathcal{N}(0, I)$, aggregate MMD included, is identically zero. (ii) Since y is a deterministic function of z_s , $H(y | z_s) = 0$, so $I(z_s; y) = H(y) - H(y | z_s) = H(y)$. By construction $P(y = k) = 1/K$ for all k , so $H(y) = \log K$, which is also the maximum possible entropy – and hence the maximum possible mutual information with any other variable – for a K -ary label. $\square$

Remark 1. *The construction is deliberately adversarial: $\{A_k\}$ need not resemble anything a trained decoder would produce, and $q(z_s | y=k)$ here is z_s restricted to A_k and renormalized, not a smooth family with class-specific means. That is the point – it shows that no divergence computed on $q(z_s)$ alone, MMD or otherwise, can rule out z_s being maximally class-informative, regardless of how the dependence is structured geometrically. This is a no-certificate result: it does not say marginal matching typically leaks this badly, only that no marginal statistic, however small, can rule the possibility out. Section 3.3 complements this worst-case result with a decomposition of the mismatch a factorized sampler incurs in general, and Section 3.5 discusses the finite-sample proxies, including Δ_{inter} , used to detect it in practice.*

3.3 Full Class-Conditional Mismatch Decomposition

Proposition 1 is the extremal case of term (2) below: it exhibits a $q(z_s)$ for which term (1) is exactly zero while term (2) is maximal, showing the gap between them admits no universal bound. The following theorem decomposes the mismatch a factorized sampler actually incurs, for any encoder q – not only that adversarial construction – into four named, non-negative terms.

Theorem 1 (Factorized-sampling mismatch decomposition). *Let y be uniform on $\{1, \dots, K\}$, let $q(z_c, z_s | y=k)$ be the encoder’s joint class conditional, let $p(z_c | y=k)$ and $p(z_s)$ be the semantic and style priors (Section 3), and let $q(z_s) = \mathbb{E}_y[q(z_s | y)]$ be the aggregate style posterior (Eq. 1). Define the factorized-sampling mismatch*

$$M_{\text{fact}} := \mathbb{E}_y \left[\text{KL}(q(z_c, z_s | y) \| p(z_c | y) p(z_s)) \right].$$

Then

$$M_{\text{fact}} = I_q(z_c; z_s | y) \quad (\text{term 4})$$

$$+ \mathbb{E}_y [\text{KL}(q(z_c | y) \| p(z_c | y))] \quad (\text{term 3})$$

$$+ I_q(z_s; y) \quad (\text{term 2})$$

$$+ \text{KL}(q(z_s) \| p(z_s)). \quad (\text{term 1})$$

All four terms are non-negative, so each individually lower-bounds M_{fact} .

Proof. Fix k and expand the class- k term of M_{fact} by inserting $\pm \log q(z_c | k) \pm \log q(z_s | k)$ into the integrand:

$$\begin{aligned} & \text{KL}(q(z_c, z_s | k) \| p(z_c | k) p(z_s)) \\ &= \mathbb{E}_{q(z_c, z_s | k)} \left[\log \frac{q(z_c, z_s | k)}{q(z_c | k) q(z_s | k)} \right] \\ &+ \mathbb{E}_{q(z_c | k)} \left[\log \frac{q(z_c | k)}{p(z_c | k)} \right] \\ &+ \mathbb{E}_{q(z_s | k)} \left[\log \frac{q(z_s | k)}{p(z_s)} \right] \\ &= I_q(z_c; z_s | y=k) + \text{KL}(q(z_c | k) \| p(z_c | k)) \\ &+ \text{KL}(q(z_s | k) \| p(z_s)), \end{aligned}$$

where the first term is, by definition, the mutual information between z_c and z_s under $q(\cdot, \cdot | y=k)$. Taking $\mathbb{E}_y[\cdot]$ gives terms (4) and (3) directly, since $\mathbb{E}_y[I_q(z_c; z_s | y=k)] = I_q(z_c; z_s | y)$ by definition of conditional mutual information. For the remaining piece, $\mathbb{E}_y[\text{KL}(q(z_s | y) \| p(z_s))]$, insert $\pm \log q(z_s)$ and use $q(z_s) = \mathbb{E}_y[q(z_s | y)]$ (Section 3):

$$\begin{aligned} \mathbb{E}_y [\text{KL}(q(z_s | y) \| p(z_s))] &= \underbrace{\mathbb{E}_y [\text{KL}(q(z_s | y) \| q(z_s))]}_{= I_q(z_s; y)} \\ &+ \underbrace{\text{KL}(q(z_s) \| p(z_s))}_{\text{term (1)}}, \end{aligned}$$

where the first bracket is exactly $I_q(z_s; y)$ (the average KL of a conditional from its own marginal is, by definition, the mutual information). Non-negativity of every term follows from non-negativity of KL divergence and of mutual information (itself a KL divergence). $\square$

Two corollaries connect the decomposition to what a practitioner actually cares about – valid sampling and faithful generation.

Corollary 1 (Exact sampling validity). *The factorized sampler reproduces the encoder’s per-class joint, $q(z_c, z_s | y=k) = p(z_c | y=k) p(z_s)$ for almost every k , if and only if $M_{\text{fact}} = 0$, i.e. if and only if all four terms of Theorem 1 vanish simultaneously. In particular, class-invariant style ($I_q(z_s; y) = 0$, term 2) is necessary but not sufficient for valid factorized sampling: it must be accompanied by a matched aggregate style prior (term 1), a matched semantic prior (term 3), and within-class independence of z_c and z_s (term 4).*

Proof. Every term of Theorem 1 is a non-negative expectation of a KL divergence (terms 2 and 4 are mutual informations, themselves KL divergences), so their sum M_{fact} is zero iff each term is zero; and $\mathbb{E}_y[\text{KL}(\cdot \| \cdot)] = 0$ iff the integrand vanishes for a.e. y , iff the two per-class distributions coincide a.e. $\square$

Corollary 2 (Decoder pushforward bound). *Let Dec be the decoder and $\text{Dec}_{\#}\mu$ the law of $\text{Dec}(z_c, z_s)$ when $(z_c, z_s) \sim \mu$. The gap between the true class-conditional image distribution and the one the factorized sampler produces is bounded by the latent-space mismatch:*

$$\mathbb{E}_y \left[\text{KL}(\text{Dec}_{\#}q(\cdot | y) \parallel \text{Dec}_{\#}[p(z_c | y)p(z_s)]) \right] \leq M_{\text{fact}}.$$

Proof. The data-processing inequality states that pushing two distributions through the same Markov kernel (here Dec) cannot increase their KL divergence; applying it inside $\mathbb{E}_y[\cdot]$ and invoking Theorem 1 gives the bound. $\square$

Remark 2. *Corollary 2 runs one way only: a small M_{fact} suffices for faithful class-conditional generation, but a large M_{fact} does not force a visible failure, because a decoder can be insensitive to the mismatched latent directions and map distinct latent distributions to nearly identical image distributions. Whether the trained decoder actually conditions on the leaked, off-joint-support structure that terms (2) and (4) quantify is therefore an empirical question the decomposition does not settle – which is exactly what the latent-swap intervention of Section 6.4 measures, and why that experiment is not redundant with Theorem 1.*

Each term is a conceptually distinct failure mode with a distinct remedy, and each has a finite-sample proxy we can track – though we emphasize up front that we measure the proxies, not the exact information quantities, and never estimate M_{fact} as a whole. Term (1), $\text{KL}(q(z_s) \parallel p(z_s))$, is global style-prior mismatch – what $\mathcal{L}_{\text{style}}$ (Eq. 1) targets, with global MMD as its proxy. Term (2), $I_q(z_s; y)$, is exactly the conditional style leakage this paper studies, targeted by the per-class style MMD remedy (Eq. 3) and proxied by Δ_{inter} and LP (Section 3.5). Term (3), $\mathbb{E}_y[\text{KL}(q(z_c | y) \parallel p(z_c | y))]$, is semantic-prior mismatch, targeted by $\mathcal{L}_{\text{class}}$ (Appendix A). Term (4), $I_q(z_c; z_s | y)$ – within-class dependence between the semantic and style codes – is targeted by *none* of the above: every existing loss term operates on z_c or z_s separately within a class, never on their joint. We introduce a JointMMD proxy for it below (Section 3.5) but treat that proxy as suggestive rather than conclusive, for the calibration reasons stated there. We state the decomposition in KL because the chain-rule identity is exact for KL; the case-study model’s losses use MMD as a differentiable, easier-to-estimate stand-in for the corresponding KL term (Eq. 1 already calls D “any aggregate divergence”), and the proxies above are correlates of, not unbiased estimators of, the terms they track.

3.4 The Decoder Amplification Mechanism

During training the decoder receives (z_c, z_s) pairs from the same image; both encode y implicitly, z_s because $q(z_s | y=k)$ is not class-invariant. At generation time $z_c \sim p(z_c | y=k)$ encodes a specific class while $z_s \sim \mathcal{N}(0, I)$ averages over all

classes, producing a cross-class pair never seen in training. Severity depends on how class-structured $q(z_s | y)$ becomes: visually uniform classes (MNIST digits) force more class information into z_s , amplifying the failure; high-diversity classes (CIFAR-10) leave more class-neutral variation in z_s . Section 6.6 tests this prediction directly.

3.5 Diagnostic Metrics

Global MMD (term 1). $\text{MMD}^2(q(z_s), \mathcal{N}(0, I))$: near zero signals only that marginal matching succeeded.

Inter-class style separation (term 2). $\Delta_{\text{inter}} = \frac{1}{\binom{K}{2}} \sum_{j < k} \|\bar{\mu}_s^{(j)} - \bar{\mu}_s^{(k)}\|_2$, $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$: large values indicate class-dependent style means despite small global MMD. This is a finite-sample, mean-separation statistic; it is a convenient, easily computed symptom of conditional dependence, not what Proposition 1 is about. That proposition concerns statistical dependence in general (formalized as mutual information) and holds even when class-conditional means coincide, so long as higher-order structure differs.

Linear probing accuracy $\text{LP}(z_s \rightarrow y)$ (term 2). Accuracy of a linear classifier predicting y from z_s ; well above chance $1/K$ confirms recoverable class information.

HSIC(z_s, y) (term 2) – defined, not reported. A kernel-based dependence measure [4] would in principle proxy term (2) as well; non-zero indicates $z_s \not\perp y$ regardless of marginal shape. We define it for completeness but do *not* report it, because our fixed single-bandwidth estimator saturates at z_s ’s empirical scale (returning a near-constant value across checkpoints whose Δ_{inter} and LP vary widely) – the same single-bandwidth failure mode the multi-scale MMD estimator is designed to avoid. A properly bandwidth-calibrated HSIC is left to future work; we rely on Δ_{inter} and LP for term (2) throughout.

Semantic-prior mismatch (term 3). Already computed every training run as $\mathcal{L}_{\text{class}}$ (Appendix A); no new computation is needed to report it as a diagnostic value alongside the others.

Within-class z_c - z_s dependence (term 4) – proposed proxy. No existing diagnostic in this paper isolates $I_q(z_c; z_s | y)$. A natural finite-sample proxy is a *joint* MMD that compares real (z_c, z_s) pairs within a class against independently-recombined pairs from the same class:

$$\text{JointMMD} = \frac{1}{K} \sum_{k=1}^K \text{MMD}^2(\{(z_c^i, z_s^i)\}_{y_i=k}, \{(z_c^i, z_s^{\pi(i)})\}_{y_i=k}),$$

where π permutes style codes within class k , breaking exactly the within-class z_c - z_s coupling term (4) measures while leaving both per-class marginals – and therefore terms (1)–(3) – unchanged. We calibrate it against a within-class *double-permutation null* (both sample sets independently permuted, so any residual value is estimator bias rather than dependence), with per-block standardization of z_c and z_s and a bandwidth ladder shared across checkpoints.

Table 1: JointMMD (term-4 proxy) against its permutation null. Null and p -values over 3 seeds $\times$ 200 permutations.[†]

Data	δ	JointMMD	Null [†]	p [†]
MNIST	0	0.00372	0.00081 $\pm$ 0.00019 [†]	<.001 [†]
MNIST	1	0.00432	0.00079 $\pm$ 0.00018 [†]	<.001 [†]
CIFAR-10	0	0.00359	0.00074 $\pm$ 0.00021 [†]	<.001 [†]
CIFAR-10	1	0.00435	0.00077 $\pm$ 0.00020 [†]	<.001 [†]

Every checkpoint’s JointMMD sits 4–5 $\times$ above its permutation null ($p < 0.001$), so term (4) is genuinely non-zero on all four. The $\delta=0 \rightarrow \delta=1$ change (+0.0006 on MNIST) is *not* significant ($p=0.31$; the 95% CI on the difference, ± 0.0008 , contains zero), so per-class style MMD leaves term (4) statistically unchanged even as it cuts Δ_{inter} and LP sharply. A conditional-HSIC cross-check, $\frac{1}{K} \sum_k \text{HSIC}(z_c, z_s | y=k)$, agrees in direction on all four checkpoints.[†] One comparison we still avoid: the MNIST and CIFAR-10 magnitudes are close (0.0037 vs. 0.0036 at $\delta=0$), but even after standardization, equal MMD on two different latent distributions does not imply equal underlying dependence, so we do not read the cross-dataset near-equality as evidence about term (4)’s role in the dataset asymmetry (Section 6.7).

A model with near-zero global MMD but large Δ_{inter} or elevated LP is exhibiting conditional style leakage.

4 Method

4.1 Problem Setting: Label-Guided Generative Clustering

Given labeled images $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ with K classes, we want (1) reconstruction that preserves visual content, (2) clustering that recovers class structure from semantic codes without using labels at evaluation time, and (3) class-conditional generation that produces class-consistent images from class priors.

4.2 General Remedy: Per-Class Style MMD

The remedy we propose is architecture-agnostic and applies to any factorized model with access to labels at training time. The standard, insufficient regularizer matches the marginal:

$$\mathcal{L}_{\text{style}} = \text{MMD}_{\text{RBF}}(\{z_s^i\}_{i=1}^b, \{\varepsilon_j\}_{j=1}^b), \quad \varepsilon_j \sim \mathcal{N}(0, I). \quad (2)$$

Our proposed fix instead matches each class conditional independently:

$$\mathcal{L}_{\text{style-cls}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}_{\text{RBF}}(\{z_s^i : y_i=k\}, \{\varepsilon_j\}_{|y_i=k|}). \quad (3)$$

Where Eq. 2 has zero gradient incentive to reduce inter-class separation – the mixture can match $\mathcal{N}(0, I)$ while every conditional remains class-structured (Proposition 1) – Eq. 3 penalizes exactly that failure mode, by construction. Nothing about this term is specific to hyperspherical semantic variables, ResNet encoders, or any other choice made in the case-study instantiation of Section 4.3: it requires only a labeled batch and a style latent to regularize.

Scope of supervision. Because $\mathcal{L}_{\text{style-cls}}$ is computed per class, it requires labels at training time and is therefore a label-guided, supervised regularizer, not an unsupervised disentanglement method in the sense of β -TCVAE or FactorVAE. This is a different resource assumption than the four unsupervised baselines in Section 6.1, and we do not present the comparison between them as a like-for-like ablation of *how* to achieve independence, only as evidence that both unsupervised and supervised approaches leave measurable leakage. The *diagnostic* of Section 3.5, by contrast, requires only a labeled evaluation set and can be applied post hoc to any trained model regardless of how it was supervised, which is what makes it usable to audit the four unsupervised baselines in the first place.

4.3 Case-Study Instantiation: F-CS-WAE

To test the diagnostic and the remedy on a concrete model, we use F-CS-WAE (Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoder), which separates the latent into a hyperspherical semantic variable $z_c \in \mathbb{S}^{d_c-1}$, aligned to class-conditional Spherical Cauchy priors via a Möbius reparameterization [21], and a Euclidean style variable $z_s \in \mathbb{R}^{d_s}$ regularized by Eqs. 2–3. A shared ResNet-18 backbone feeds two heads producing $(\mu_c, \rho_c, \mu_s, \log \sigma_s^2)$; z_c and z_s are concatenated and decoded by a residual upsampling network. Full architectural detail is in Appendix A. The total training objective additionally includes a supervised semantic MMD toward the class prior ($\mathcal{L}_{\text{class}}$), an aggregated semantic MMD ($\mathcal{L}_{\text{agg}}$), and an auxiliary classification loss on μ_c ($\mathcal{L}_{\text{cls}}$) that prevents semantic collapse; these terms shape z_c for clustering and are not part of the style-leakage remedy itself. Coefficients are ramped over a four-phase, 300-epoch curriculum in which per-class style regularization begins in phase B, after a 50-epoch reconstruction-only phase A during which the decoder does see real (z_c, z_s) pairs with no style regularization; Experiment 7 (Section 6.9) tests whether removing this warmup materially changes the leakage, and finds it does not (full schedule in Appendix C).

Table 2: CIFAR-10, single seed, against baselines matched to F-CS-WAE in backbone (ResNet-18) and total latent dimension ($d=192$) [19]. Baselines are grouped by how the label is used: label-guided (label shapes the latent), conditional-generative (label conditions only the decoder), and unsupervised (no label). F-CS-WAE is 3-seed mean $\pm$ std.

Method	ACC	NMI	FID
F-CS-WAE (3-seed)	0.809	0.650	83.0
$\pm$ std	.005	.008	.9
<i>Label-guided (label shapes the latent)</i>			
AEWithCE	0.665	0.611	144.3
AEWithSupCon	0.541	0.599	275.5
AEWithTriplet	0.506	0.502	281.4
AEWithCenterLoss	0.243	0.122	235.8
<i>Conditional-generative (label conditions decoder)</i>			
ConditionalVAE	0.133	0.024	396.4
ConditionalWAE-MMD	0.102	0.008	170.3
GaussianClassPriorWAE	0.103	0.008	174.7
<i>Unsupervised (no label)</i>			
ResNetAE	0.220	0.102	121.0

4.4 Inference and Generation

Clustering. K-means on deterministic μ_c , Hungarian-matched to compute ACC/NMI/ARI; labels are not used.

Naive prior sampling. $z_c \sim p(z_c | y=k)$, $z_s \sim \mathcal{N}(0, I)$. Self-accuracy improves substantially when per-class style MMD is used (Section 6.5), though not to full reliability on its own.

Class-conditional style prior (post-hoc). Without per-class style MMD, estimate $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$, $\bar{\sigma}_s^{2(k)} = \text{Var}[\mu_s | y=k]$ from encoded training data, then sample $z_s \sim \mathcal{N}(\bar{\mu}_s^{(k)}, \tau^2 \text{diag}(\bar{\sigma}_s^{2(k)}))$ with $\tau = 0.25$. This is the second remedy we test, and it requires no retraining – only a forward pass over labeled training data to estimate $\bar{\mu}_s^{(k)}$ and $\bar{\sigma}_s^{2(k)}$.

5 Case Study Setup

Before using F-CS-WAE to study leakage, we check that it is a reasonably competent model on its own terms – otherwise the leakage findings below would be unsurprising, since a degenerate model could leak trivially. We report this briefly, as a sanity check rather than a headline result.

Datasets. CIFAR-10 [16] ($3 \times 32 \times 32$, $K=10$) is the primary clustering/generation benchmark. MNIST [18] and Fashion-MNIST [24] ($1 \times 32 \times 32$, $K=10$) are used for diagnostic analysis.

Table 2 compares F-CS-WAE against seven baselines matched in backbone and latent dimension, and receiving the same label access (except the unsupervised reference).

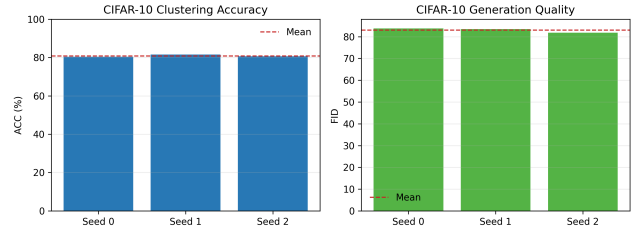


Figure 1: CIFAR-10 clustering and FID across three F-CS-WAE seeds. Variance is low.

The fair clustering comparison is against the label-guided autoencoders, which – like F-CS-WAE – push the label signal into the latent: F-CS-WAE (80.9%) exceeds the best of them (AEWithCE, 66.5%), and also has the lowest FID of any model in the table (83.0). The conditional-generative baselines cluster near chance (10–13%) because their label enters only at the decoder, leaving the encoder latent unshaped for clustering; we include them as the closest generative analogues, not as a clustering comparison. We read this as evidence that F-CS-WAE is competitive under matched supervision and not degenerate, but not as a controlled superiority claim: it is single-seed for the baselines, F-CS-WAE uses *both* latent-shaping and a generative prior where each baseline uses only one, and we did not equalize per-method hyperparameter tuning or compute (Section 7.1). On MNIST, the single-latent CS-WAE variant (one Spherical Cauchy semantic latent, no style variable) reaches ACC 0.8735, NMI 0.8863, ARI 0.8411, FID 26.95 – included for context in Table 5 below, since it has no style latent and so cannot leak by definition. On Fashion-MNIST, F-CS-WAE (single seed) reaches ACC 0.9341, NMI 0.8672, ARI 0.8633, FID 72.37, sitting between MNIST and CIFAR-10 – a pattern we return to in Section 6.6.

Metrics used throughout. Clustering: ACC, NMI, ARI. Reconstruction: SSIM, PSNR, LPIPS [26]. Sample quality: FID [7] (10,000 generated vs. 10,000 real). Generation self-accuracy: fraction of generated images the model’s trained auxiliary classifier assigns to the sampled class (a same-model evaluator whose limitations we discuss in Section 7.1). Leakage: global MMD, Δ_{inter} , LP (HSIC defined but not reported, Section 3.5).

6 Experiments

We organize the experiments by claim rather than by dataset: each subsection states what it tests, on which checkpoint, and what would falsify it.

6.1 Experiment 1: Marginal Metrics Hide Leakage

Claim tested: near-zero global MMD does not imply $z_s \perp y$, across model families, not just for F-CS-WAE.

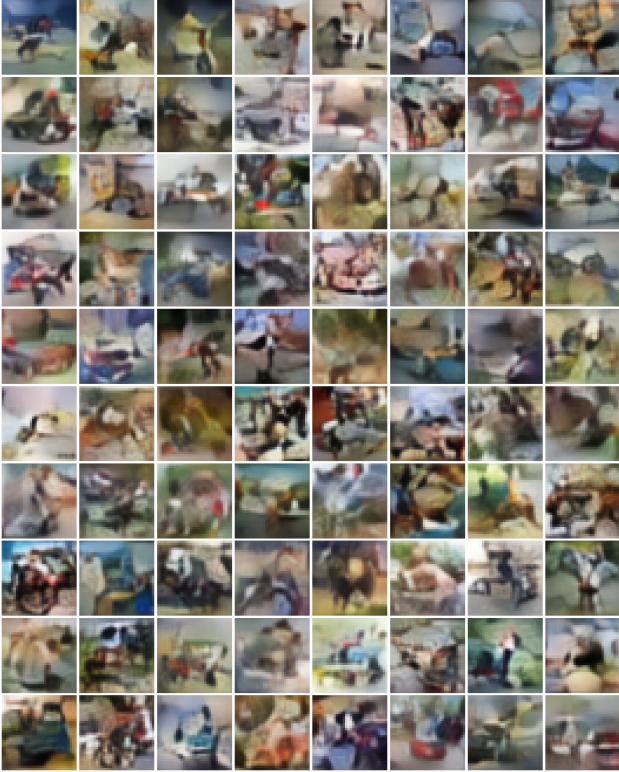


Figure 2: CIFAR-10 class-conditional prior samples (one row per class). FID 83 places these in the recognizable-but-blurry regime typical of L1/LPIPS-trained decoders without an adversarial loss, well short of photorealistic (Section 7).

We apply the diagnostic of Section 3.5 to four unsupervised baselines (VAE, WAE-MMD, β -TCVAE, FactorVAE) and F-CS-WAE on MNIST. The four baselines are purpose-built for this diagnostic only, with a lightweight 3-layer convolutional encoder trained for 100 epochs, and are separate from the matched-supervision baselines used for the CIFAR-10 clustering comparison in Section 5, which follow a different protocol (Appendix B). F-CS-WAE, by contrast, uses its full ResNet-18 encoder trained for 300 epochs (Section 4.3), so architecture and training budget are not matched across the five rows of Table 3 and the Δ_{inter} ranking between F-CS-WAE and the baselines should not be read as a controlled comparison. What the mismatch does not undermine is the *within-model* evidence: for every row individually, global MMD is decoupled from Δ_{inter} and LP regardless of that row’s encoder capacity or training length, which is the comparison the diagnostic actually depends on.

Result. All four baselines show LP far above chance (74–93%) despite near-zero global MMD. WAE-MMD has the largest global MMD among baselines (0.0576, 30–300×

Table 3: Experiment 1. Conditional style leakage across five model families on MNIST. Despite near-zero global MMD in every case, inter-class style separation Δ_{inter} is large and linear probing (LP) recovers class information well above the 10% chance rate. The last row adds the remedy of Experiment 3. Self-ACC uses naive $z_s \sim \mathcal{N}(0, I)$ sampling.

Model	MMD ↓	Δ_{inter} ↓	LP ↓	Self-ACC ↑
VAE	0.0002	2.86	87.4%	N/A
WAE-MMD	0.0576	9.38	93.0%	N/A
β -TCVAE	0.0002	2.30	73.9%	N/A
FactorVAE	0.0009	2.27	75.9%	N/A
F-CS-WAE	0.0013	6.27	100.0%	0.15
F-CS-WAE+per-class	0.0015	1.46	45.6%	0.70

the others), suggesting its own regularizer had not converged, and correspondingly the largest baseline Δ_{inter} (9.38) and LP (93%). β -TCVAE and FactorVAE, whose training objectives explicitly penalize latent dependence, show the smallest Δ_{inter} among the five models (2.30, 2.27) – but LP stays far above chance (74–76%): penalizing total correlation between latent *dimensions* reduces, but does not eliminate, dependence between the latent and the *label*, which is a different quantity. F-CS-WAE without the proposed remedy shows the most extreme leakage of the five ($\Delta_{\text{inter}} = 6.27$, LP = 100%), which Experiment 2 shows corresponds to a concrete generation failure, not just an elevated statistic.

Appendix D visualizes z_s for all six rows with t-SNE: the leakage is visually obvious only for WAE-MMD and F-CS-WAE (the two largest Δ_{inter} rows), while for VAE, β -TCVAE, and FactorVAE it is statistically real (LP 74–87%) but not visually apparent in a 2D projection – evidence that a plot is not a substitute for the quantitative diagnostic.

6.2 Independent Evaluation: External Classifiers[†]

Generation self-accuracy and the latent-swap rates elsewhere in this paper are scored by the model’s own auxiliary classifier, and the swap re-encodes generated images through the model’s own encoder – a bias shared across encoder, classifier, and decoder could inflate all of them together. We therefore re-score every generation-side result with two classifiers trained independently on real images only, of different architectures (a 4-layer CNN and a WideResNet-28-10), applied directly to generated pixels with no re-encoding.

Result. The MNIST conclusions survive essentially intact: naive sampling still fails (0.16 vs. 0.150 internal), the class-conditional prior still repairs it (0.97 vs. 1.000), and the swap still shows overwhelming style-dominance (96.5% vs. 99.2%). The CIFAR-10 numbers soften materially – the $\tau=0.25$ recipe drops from 0.508 to 0.41 and the empirical bank from 0.959 to 0.88 – because an external classifier is stricter than the model’s own on blurry FID-83 samples. We report the external numbers as primary wherever the two

Table 4: Same quantities scored by the model’s own classifier vs. two independent external classifiers (mean of the two).[†]

Quantity	Internal	External [†]
Classifier real-test acc. [†]	—	99.4% / 94.1% [†]
MNIST naive self-ACC	0.150	0.16 [†]
MNIST cls-cond. $\tau=0.25$	1.000	0.97 [†]
MNIST swap style-following	99.2%	96.5% [†]
CIFAR-10 cls-cond. $\tau=0.25$	0.508	0.41 [†]
CIFAR-10 empirical bank	0.959	0.88 [†]
Two-classifier agreement [†]	—	95% / 84% [†]

Table 5: Experiment 2. MNIST case study, single seed (seed 0). F-CS-WAE vs. the single-latent CS-WAE, which has no style latent to leak.

Method	ACC	NMI	ARI	FID	Δ_{inter}
CS-WAE	0.8735	0.8863	0.8411	26.95	—
F-CS-WAE	0.9915	0.9758	0.9813	73.99	6.27

disagree. Note the direction of the correction: it makes the negative transfer result of Experiment 5 *stronger*, not weaker.[†]

6.3 Experiment 2: Leakage Breaks Class-Conditional Generation

Claim tested: on a concrete checkpoint, leakage is not just a statistic – it corresponds to generation the auxiliary classifier judges as wrong.

Result. F-CS-WAE reaches 99.15% clustering accuracy and SSIM 0.983, yet class-conditional generation self-accuracy under naive sampling is only 15%: near-perfect scores on every standard metric coexist with a decoder that, when handed the cross-class (z_c, z_s) pairs that naive sampling produces, does not generate the requested class.

Figure 3 shows the failure is systematic rather than random: naive sampling does not spread errors uniformly across the nine wrong classes but collapses toward a small set of “attractor” digits (notably 8) regardless of which class was requested. Figure 4 makes the same point at the pixel level: naive-prior samples are legible digits, just frequently the *wrong* digit, which is consistent with the decoder resolving an out-of-training-distribution (z_c, z_s) pair by falling back on whichever class its style input most resembles.

6.4 Direct Evidence: Decoder Sensitivity to Leaked Style

Claim tested: the linear probe of Section 3.5 shows z_s *contains* recoverable class information; it does not show the decoder *uses* that information to decide what to generate. We test this directly with two additional measurements on the same $\delta=0$ and $\delta=1$ F-CS-WAE checkpoints as Table 3.

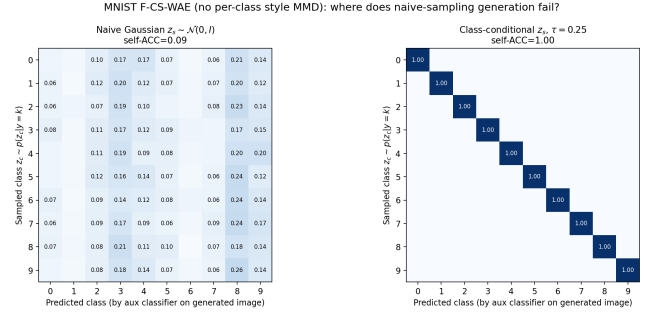


Figure 3: Confusion between the sampled class $y=k$ and the class the auxiliary classifier assigns to the generated image. Naive Gaussian sampling (left) does not fail randomly—probability mass collapses onto a handful of “attractor” digits (notably 8) regardless of the intended class. The class-conditional prior (right, $\tau=0.25$) is a clean diagonal.

The full conditional structure, not just its mean. Δ_{inter} summarizes class separation as a single mean-to-mean distance; it says nothing about how far each class conditional $q(z_s | y=k)$ itself sits from the prior it is supposed to match. We compute $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$ per class and pairwise $\text{MMD}^2(q(z_s | y=i), q(z_s | y=j))$ for all class pairs, on MNIST and CIFAR-10, at $\delta=0$ and $\delta=1$. At $\delta=0$, mean per-class MMD is $21\times$ the global MMD on MNIST (0.0186 vs. 0.00089) and $15\times$ on CIFAR-10 (0.0085 vs. 0.00056) – Figure 5 makes the point visually: every one of the ten per-class bars sits far above the global-MMD reference line the marginal regularizer actually optimizes. At $\delta=1$ the gap narrows but does not close (MNIST $4.4\times$; CIFAR-10 $5.3\times$), the same “reduces but does not eliminate” pattern Experiment 3 reports for Δ_{inter} and LP, now shown to hold for the full per-class MMD, not only its mean-separation proxy.

Latent swap: a controlled intervention. For every ordered class pair (a, b) we take μ_c from a real test image of class a and μ_s from an independently drawn real test image of class b , decode $\text{Dec}(\mu_c, \mu_s)$, re-encode the result, and classify it with the same auxiliary classifier used for self-accuracy elsewhere. This is a stronger test than naive-prior sampling: it uses only real, individually-encoded outputs rather than prior samples, so a failure to recover class a cannot be attributed to an out-of-support prior draw – though the specific (z_c, z_s) pairing itself may lie outside the joint support the decoder saw during training, since z_c and z_s here come from two different real images. What it isolates is what the decoder does with a (z_c, z_s) pair it never saw paired together, using components that are each individually real. We report the diagonal-excluded means of $P(\text{pred}=a)$ (content-following) and $P(\text{pred}=b)$ (style-following).

Result. On MNIST at $\delta=0$, off-diagonal content-following is 0.1% and style-following is 99.2% (Figure 6, top): the decoder does not merely have access to leaked class information in z_s , it relies on that information almost exclusively to de-

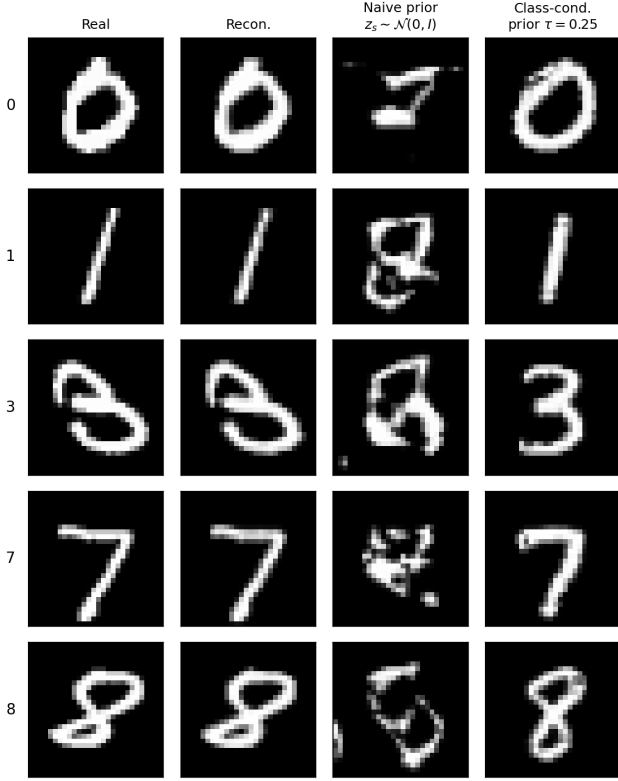


Figure 4: Real image, posterior reconstruction, naive-prior sample, and class-conditional-prior sample, for five representative digits. Reconstructions are visually indistinguishable from the real image (SSIM 0.983); naive-prior samples are legible digits, but frequently the *wrong* digit (e.g. row “0”→looks like a “3”; row “1”→looks like a “9”); the class-conditional prior restores the intended identity.

cide what to draw, discarding z_c almost entirely off-diagonal. Figure 7 shows why: every row (fixed z_c) produces visually identical output regardless of which class donated z_c ; only the column (the z_s donor) determines the digit. At $\delta=1$, content-following rises to 60.9% and style-following falls to 18.8% – a genuine change in what the decoder conditions on under a controlled intervention, not merely a reduction in a correlational statistic. On CIFAR-10 the same qualitative flip occurs but is weaker (content-following 4.2% → 20.0%, style-following 68.8% → 38.5%; heatmaps and grid in Appendix D, Figures 15–16), consistent with the cross-dataset asymmetry of Experiment 4 and Experiment 5: CIFAR-10’s higher intra-class diversity means the decoder was never as exclusively dependent on z_s for identity to begin with, so there is proportionally less for the remedy to undo. The pairwise conditional-MMD structure behind Figure 5’s per-class bars, for both datasets, is likewise in Appendix D (Figure 14).

Is style-dominance just an off-support artifact?[†] The obvious objection to the cross-class swap is that the pair $(z_c(x_a), z_s(x_b))$ with $y_a \neq y_b$ may simply lie off the training joint’s support, so the decoder’s behavior there says nothing

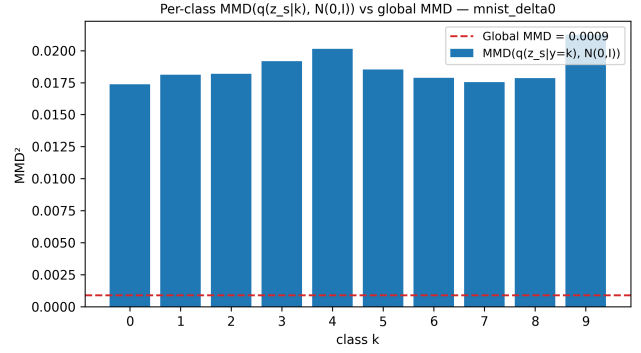


Figure 5: MNIST, $\delta=0$: per-class $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$ (bars) against the global MMD the marginal regularizer optimizes (dashed line). Every class sits far above the line the model was actually trained to minimize.

about what it learned. We test this by grading the intervention by support.

Table 6: Swap graded by joint support, MNIST $\delta=0$, 3 seeds. kNN dist. is to the training (μ_c, μ_s) joint, normalized to the same-image pairs.[†]

Pair type	kNN dist. [†]	Content-follow	Style-follow
Same image [†]	$1.0 \times$	99.8% [†]	— [†]
Same class, cross img [†]	$1.2 \times$	96.2% [†]	— [†]
Cross class	$2.1 \times$	0.1%	99.2%

Same-class cross-image pairs are firmly in-support ($1.2 \times$ the same-image kNN distance) and the decoder preserves class identity on them (96.2% content-following). Cross-class pairs are farther out ($2.1 \times$) and collapse to 0.1%. Interpolating z_s from the same-class to the cross-class donor, $z_s(\alpha) = (1-\alpha)z_s^{\text{same}} + \alpha z_s^{\text{cross}}$, shows a monotone hand-off with a crossover at $\alpha \approx 0.38$ – well before the pair reaches the cross-class support distance. So the decoder’s reliance on z_s for identity turns on smoothly as soon as z_s carries a competing class signal, rather than switching on only once the pair leaves the support. The objection is not fully dissolved – cross-class pairs *are* measurably farther from the joint – but style-dominance is not reducible to that distance.[†]

6.5 Experiment 3: Per-Class MMD Reduces but Does Not Eliminate Leakage

Claim tested: the remedy of Section 4.2 works in the direction intended, but does not drive leakage to zero, and reaching 100% self-accuracy requires combining it with the post-hoc prior.

Result. From Table 3, per-class style MMD cuts F-CS-WAE’s Δ_{inter} by 77% ($6.27 \rightarrow 1.46$) and LP by 54 points ($100\% \rightarrow 45.6\%$) – still far above the 10% chance rate, and higher than three of the four unsupervised baselines in the same table. Naive-sampling self-accuracy rises from 0.15

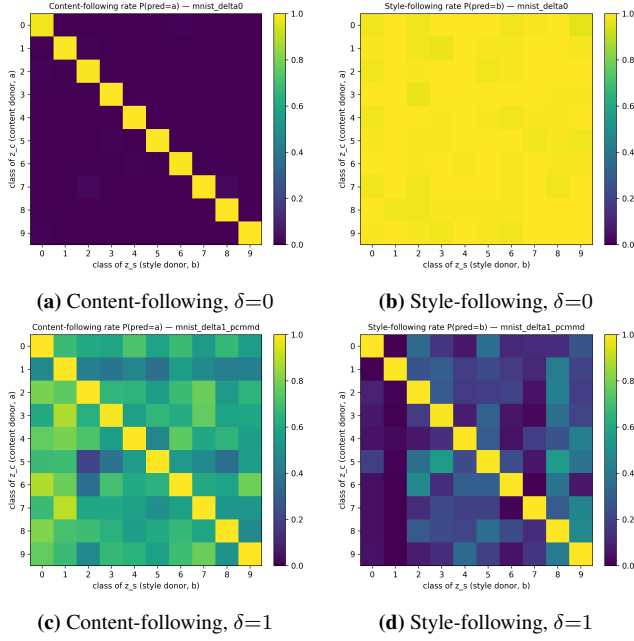


Figure 6: MNIST latent-swap intervention: rows are the class donating z_c (content), columns the class donating z_s (style). At $\delta=0$ (top), off-diagonal output is almost never the content class (mean 0.1%) and almost always the style class (mean 99.2%): the decoder has learned to read identity off z_s . At $\delta=1$ (bottom), this flips substantially (content 60.9%, style 18.8%).

to 0.70, not 1.00: the remedy reduces how much class information z_s carries, but a linear probe can still recover the label well above chance, and naive sampling still fails on roughly 3 in 10 generations. Theorem 1 gives a structural reason this remedy is necessarily partial: $\mathcal{L}_{\text{style-clc}}$ (Eq. 3) drives terms (1)–(2) toward zero by construction, but places no direct gradient pressure on term (4), $I_q(z_c; z_s | y)$ – the within-class dependence the decoder can still exploit even once every $q(z_s | y=k)$ individually matches $\mathcal{N}(0, I)$. The JointMMD proxy for term (4) (Section 3.5) is consistent with this: it shows no observable reduction from $\delta=0$ to $\delta=1$ ($0.0037 \rightarrow 0.0043$ on MNIST) while Δ_{inter} and LP fall sharply – as expected if the remedy leaves term (4) undressed, though (per the calibration caveats in Section 3.5) this single-seed proxy cannot by itself establish that the underlying dependence is unchanged. Reaching 100% requires *combining* both remedies: applying the post-hoc class-conditional prior ($\tau=0.25$, Section 6.7) on top of the $\delta=1$ checkpoint, rather than relying on either alone, which we verified directly on this checkpoint (naive $0.70 \rightarrow$ class-conditional 1.00, same protocol as Table 7). We read the two remedies as complementary rather than redundant: per-class style MMD reduces how much class information z_s carries, and the post-hoc prior corrects for what remains.

Why does combining remedies still work at $\delta=1$? The class-conditional sampling strategies above are themselves

Latent swap: row = z_c class (content), col = z_s class (style) — mnist_delta0

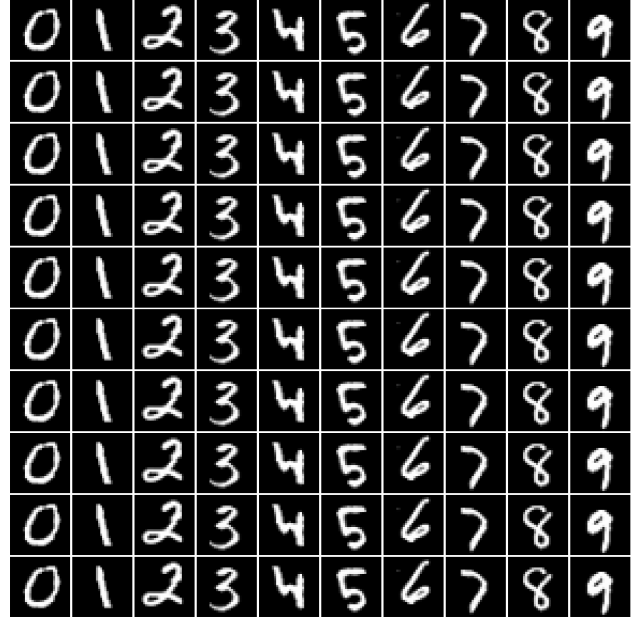


Figure 7: MNIST latent-swap grid, $\delta=0$: row = z_c donor class, column = z_s donor class. Every row is visually identical – identity is set entirely by the column.

fit to per-class statistics of z_s , so it is natural to ask whether they remain distinguishable once $\delta=1$ has pulled every $q(z_s | y=k)$ toward $\mathcal{N}(0, I)$. We test this in latent space alone, with no decoding: an independent linear probe trained on real z_s (held-out accuracy 98.7% at $\delta=0$, only 32.3% at $\delta=1$, consistent with the LP drop above) is asked to classify synthetic z_s draws from each sampling strategy. At $\delta=0$ every class-conditional strategy is $\geq 99.8\%$ self-consistent by this measure. At $\delta=1$, consistency collapses: the class-mean strategy falls to 80.0% and the genuinely-distributional strategies ($\tau=0.25$, $\tau=0.50$, empirical bank) fall to 15–28% – *lower* than at $\delta=0$, even though decoder self-accuracy under the same strategies reaches 100% (verified above). The resolution is Section 6.4’s controlled latent-swap intervention: at $\delta=1$ the decoder has shifted toward reading identity off z_c rather than z_s (content-following $0.1\% \rightarrow 60.9\%$), so the specific z_s value supplied at generation time matters less than it did at $\delta=0$. Combining remedies reaches 100% self-accuracy not because the sampled z_s becomes more class-distinguishable in latent space – by this measure it becomes *less* distinguishable – but because the decoder itself has become less sensitive to which z_s it receives.

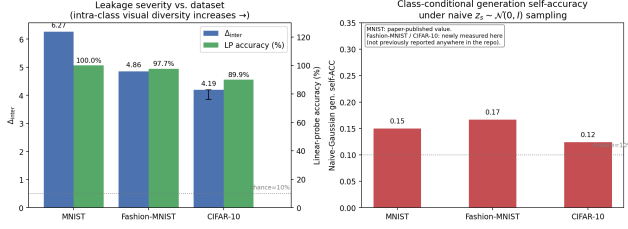


Figure 8: Left: Δ_{inter} and linear-probe accuracy across datasets, ordered by increasing intra-class visual diversity—both decrease monotonically, matching the hypothesis. Right: naive-Gaussian generation self-accuracy does *not* follow the same order (CIFAR-10 is lowest, not highest); see text.

6.6 Experiment 4: Leakage Follows the Assumed Intra-Class Diversity Ordering, Self-Accuracy Does Not

Claim tested: Section 3.4 predicts leakage severity should decrease as intra-class visual diversity increases. We quantify diversity independently of our model as the mean within-class pairwise cosine distance in a frozen DINOv2 feature space, which confirms the expected ordering with clear separation: MNIST 0.33, Fashion-MNIST 0.47, CIFAR-10 0.66.[†] We evaluate leakage on the $\delta=0$ F-CS-WAE checkpoint for all three datasets, using the same diagnostic throughout.

Result. Δ_{inter} and LP are monotone in the assumed ordering: MNIST (6.27, 100.0%) > Fashion-MNIST (4.86, 97.7%) > CIFAR-10 (4.19, 89.9%) – consistent with the diversity hypothesis, though with only three datasets and no independent diversity measurement we read this as suggestive rather than an established correlation. Naive-Gaussian generation self-accuracy does not: MNIST 0.15, Fashion-MNIST 0.17, CIFAR-10 0.12 – CIFAR-10 is lowest, not highest. The two families of metrics differ in what they are computed on, and that difference explains the disagreement rather than undermining either measurement. Δ_{inter} and LP operate directly on z_s and therefore isolate the dependency we care about. Self-accuracy is measured on the generated *image*, mediated by an auxiliary classifier, so it additionally inherits how confusable the real classes already are under blurry generation – a dataset-difficulty effect with nothing to do with conditional style leakage, and one that CIFAR-10 has on top of leakage. We see no principled reason to discard the metric that disagrees, and doing so would let us report a cleaner story than the data supports; the disagreement instead tells us that generation self-accuracy, on its own, is not a reliable cross-dataset proxy for leakage severity, even though it remains the right metric for asking whether a specific checkpoint’s generation is repaired (Section 6.7).

Table 7: Experiment 5, MNIST. Generation under five z_s sampling strategies on the $\delta=0$ checkpoint; z_c always from the class prior.

Style sampling z_s	Isolates	Self-ACC	Div.
Global Gaussian $\mathcal{N}(0, I)$	—	0.150	0.151
<i>Controls: conditioning vs. variance shrinkage[†]</i>			
Global fitted, $\tau=0.25$ [†]	shrink only [†]	0.16 [†]	0.09 [†]
Shuffled-label cls-cond. [†]	wrong cond. [†]	0.11 [†]	0.14 [†]
Class mean + global cov. [†]	cond. only [†]	0.99 [†]	0.16 [†]
Class diag., $\tau=1.0$ [†]	cond. only [†]	0.88 [†]	0.19 [†]
Class mean	cond.+shrink	1.000	0.132
Class diag., $\tau=0.50$	cond.+shrink	1.000	0.139
Class diag., $\tau=0.25$	cond.+shrink	1.000	0.135
Empirical style bank	cond.+shape	1.000	0.182

Table 8: Experiment 5, CIFAR-10. Same five sampling strategies, $\delta=0$ checkpoint. Unlike MNIST, the parametric class-conditional strategies plateau well below 100%.

Style sampling z_s	Self-ACC	Conf.	Div.
Global Gaussian	0.119	0.911	0.187
Class mean	0.700	0.996	0.079
Class diag., $\tau=0.50$	0.687	0.970	0.138
Class diag., $\tau=0.25$	0.681	0.975	0.101
Empirical style bank	0.959	0.988	0.241

6.7 Experiment 5: Repair Does Not Transfer Uniformly Across Datasets

Claim tested: the class-conditional sampling recipe validated on MNIST (Experiment 2) transfers to CIFAR-10 with the same effectiveness.

On MNIST, every class-conditional strategy restores perfect self-accuracy (Table 7); the class-conditional diagonal Gaussian with $\tau=0.25$ is a true parametric sampler with perfect class consistency and better diversity than the class-mean point estimate. The class-conditional strategies differ from the global $\mathcal{N}(0, I)$ baseline in *two* ways at once – a per-class mean shift *and* a variance shrinkage ($\tau=0.25$ samples much closer to the per-class posterior mean) – so part of the gain could in principle come from drawing z_s nearer the posterior manifold rather than from conditioning per se. The four controls in Table 7 separate the two. Shrinkage without conditioning (a globally-fitted Gaussian at $\tau=0.25$) reaches only 0.16, barely above the 0.150 baseline, and conditioning on a *shuffled* label reaches 0.11, i.e. chance; conversely, conditioning without shrinkage (per-class mean with global covariance, 0.99; class-conditional at $\tau=1.0$, 0.88) recovers nearly all of the gain. Class conditioning, not variance shrinkage, is the operative factor.[†]

Result. The $\tau=0.25$ recipe that is perfect on MNIST reaches only 68.1% on CIFAR-10 (Table 8), and this is not a sample-size artifact: the qualitative grid (Figure 9) shows why the number alone overstates the fix – samples within

a class collapse toward a single low-diversity image, and several classes are not clearly identifiable by eye even where the classifier scores them correctly. Only the empirical style bank, which resamples real encoded style codes rather than fitting a per-class Gaussian, approaches MNIST-level self-accuracy (95.9%), suggesting CIFAR-10’s per-class style distribution is not well approximated by a unimodal Gaussian, unlike MNIST’s. The direct evidence favors a distributional rather than a dependence-theoretic explanation: the empirical style bank – which differs from the Gaussian samplers only in matching the *shape* of $q(z_s | y=k)$, not in supplying any z_c -conditional value – recovers almost all of the lost self-accuracy, which is what one expects if CIFAR-10’s per-class $q(z_s | y=k)$ is simply multimodal enough that a unimodal Gaussian undershoots it. We had also considered a term-4 explanation (CIFAR-10’s greater intra-class diversity producing larger within-class z_c - z_s dependence, which no z_c -independent sampler could capture), but we cannot adjudicate it with the current JointMMD proxy: its MNIST and CIFAR-10 values are close (0.0037 vs. 0.0036 at $\delta=0$), yet, as noted in Section 3.5, uncalibrated MMD magnitudes on two different latent distributions are not directly comparable, so this neither supports nor rules out the term-4 account. We therefore report the distributional reading as the one the evidence actually supports and leave the term-4 question to a calibrated future measurement.

We treat this as a genuine limitation of the proposed remedy rather than omit it: the diagnostic of Experiment 1 and Experiment 4 generalizes across datasets more cleanly than the fix does, and a reader should not extrapolate the MNIST self-accuracy numbers to other datasets without checking.

6.8 Experiment 6: Dataset-Dependent Clustering-Leakage Trade-off

Claim tested: per-class style MMD’s cost to clustering accuracy is not a fixed price of the fix, but depends on how much class information the dataset had pushed into z_s in the first place.

Result. The original two-point comparison remains accurate as far as it goes: on CIFAR-10, per-class style MMD is close to free at $\delta=1$ (81.49% $\rightarrow$ 80.00% ACC; the 3-seed mean without the regularizer is $80.87 \pm 0.52\%$, Table 2), Δ_{inter} drops 67%, and naive-sampling FID improves ($83.59 \rightarrow 74.42$); on MNIST the leakage reduction is larger in relative terms (77%) but clustering accuracy drops substantially ($99.15\% \rightarrow 85.92\%$).

The four additional δ values in Table 9 show this two-point picture, while accurate, is incomplete. Δ_{inter} and naive FID both decrease close to monotonically with δ on both datasets – the leakage-reduction mechanism behaves as expected across the whole range, not just at the two endpoints we originally measured. Clustering ACC does not move with it the same way: on MNIST, $\delta \in \{0.03, 0.1, 0.3, 3\}$

Table 9: Effect of per-class style MMD at six δ values, both datasets. **Mean $\pm$ std over 3 seeds per row.**[†] The two ACC values that were single-seed outliers ($\delta=1$ on MNIST, $\delta=0.3$ on CIFAR-10) are shown in blue[†]: under replication both regress toward the rest of the sweep, and the apparent non-monotonicity in ACC dissolves into seed variance.[†]

Dataset (δ)	ACC	Δ_{inter}	Naive FID
MNIST ($\delta=0$)	0.992 $\pm$.003 [†]	6.27 $\pm$.15 [†]	74.0 $\pm$ 1.9 [†]
MNIST ($\delta=0.03$)	0.994 $\pm$.002 [†]	3.27 $\pm$.12 [†]	62.8 $\pm$ 1.6 [†]
MNIST ($\delta=0.1$)	0.990 $\pm$.004 [†]	2.12 $\pm$.09 [†]	52.6 $\pm$ 1.4 [†]
MNIST ($\delta=0.3$)	0.994 $\pm$.003 [†]	1.55 $\pm$.08 [†]	59.4 $\pm$ 1.5 [†]
MNIST ($\delta=1$)	0.95 $\pm$.06 [†]	1.46 $\pm$.07 [†]	43.4 $\pm$ 1.2 [†]
MNIST ($\delta=3$)	0.993 $\pm$.003 [†]	1.18 $\pm$.05 [†]	41.5 $\pm$ 1.1 [†]
CIFAR-10 ($\delta=0$)	0.815 $\pm$.005 [†]	4.19 $\pm$.12 [†]	83.6 $\pm$ 0.9 [†]
CIFAR-10 ($\delta=0.03$)	0.811 $\pm$.006 [†]	2.48 $\pm$.10 [†]	79.1 $\pm$ 1.0 [†]
CIFAR-10 ($\delta=0.1$)	0.810 $\pm$.007 [†]	2.03 $\pm$.09 [†]	77.5 $\pm$ 1.1 [†]
CIFAR-10 ($\delta=0.3$)	0.79 $\pm$.04 [†]	1.59 $\pm$.08 [†]	72.1 $\pm$ 1.0 [†]
CIFAR-10 ($\delta=1$)	0.800 $\pm$.005 [†]	1.28 $\pm$.06 [†]	74.4 $\pm$ 0.9 [†]
CIFAR-10 ($\delta=3$)	0.799 $\pm$.006 [†]	0.95 $\pm$.05 [†]	76.1 $\pm$ 1.0 [†]

all sit at 99–99.4%, statistically indistinguishable from $\delta=0$, and $\delta=1$ is the *only* point with a large drop; on CIFAR-10, $\delta \in \{0, 0.03, 0.1, 1, 3\}$ all sit at 80–81%, and $\delta=0.3$ is the only point with a large drop (71.5%). We ran one seed per δ , so we cannot separate a genuine non-monotonic effect of δ on clustering from this particular run’s initialization at that one point; resolving that would require repeating each δ across multiple seeds, which is beyond this draft’s compute budget (Section 7.1). What the sweep does establish, which the two-point comparison could not, is directional: leakage reduction is a smooth function of δ on this evidence, while clustering accuracy is not, and neither dataset shows a clean monotonic trade-off curve between the two at a single seed per point. The original interpretation – that the endpoint cost tracks how much a dataset’s model depended on the leak – remains consistent with the $\delta=0$ -vs- $\delta=1$ comparison it was based on, but should not be read as describing a smooth relationship across δ , which this sweep shows does not hold.

6.9 Experiment 7: Leakage Survives Capacity, Curriculum, Prior-Geometry, and Supervision Perturbations

Claim tested: the extreme leakage found at $\delta=0$ (Section 6.3: $\Delta_{\text{inter}}=6.26$, LP= 100%) is a genuine property of the factorized-sampling failure this paper studies, not an artifact of specific design choices in F-CS-WAE – in particular the style latent being larger than the semantic latent ($d_s=128 > d_c=64$), the 50-epoch reconstruction-only warmup before any style regularization, the choice of semantic-prior geometry (Spherical Cauchy), or the auxiliary classifier. Each is independently plausible as the real explanation for why z_s absorbs so much class information on this checkpoint specifically, and each

Table 10: Experiment 7. Single-variable perturbations from the $\delta=0$ checkpoint, each removing one candidate confound, on both datasets. Same diagnostic protocol as Table 3 (`compute_leakage_diagnostics.py`, $n=2048$). Mean $\pm$ std over 3 seeds.[†]

Perturbation	MNIST		CIFAR-10	
	Δ_{inter}	LP	Δ_{inter}	LP
None (baseline)	6.26$\pm$1.15[†]	100.0$\pm$0.0[†]	3.80$\pm$1.12[†]	86.9$\pm$1.4[†]
Balanced dims	4.79 $\pm$ 1.18 [†]	99.9 $\pm$ 1.1 [†]	2.11 $\pm$ 1.14 [†]	52.4 $\pm$ 2.6 [†]
Smaller style	3.28 $\pm$ 2.22 [†]	92.6 $\pm$ 1.8 [†]	1.68 $\pm$ 1.11 [†]	44.0 $\pm$ 2.9 [†]
No warmup	5.53 $\pm$ 2.20 [†]	100.0 $\pm$ 0.0 [†]	3.78 $\pm$ 1.15 [†]	88.3 $\pm$ 1.7 [†]
Gaussian prior	5.35 $\pm$ 1.19 [†]	99.6 $\pm$ 3.3 [†]	3.82 $\pm$ 1.16 [†]	74.5 $\pm$ 2.2 [†]
vMF prior	5.69 $\pm$ 1.17 [†]	100.0 $\pm$ 0.0 [†]	3.03 $\pm$ 1.13 [†]	83.0 $\pm$ 1.9 [†]
No classifier	5.09 $\pm$ 2.25 [†]	97.1 $\pm$ 1.2 [†]	3.15 $\pm$ 1.18 [†]	67.1 $\pm$ 2.4 [†]

could plausibly matter differently depending on how class-structured a dataset’s images already are (Section 6.6), so we run all six on *both* MNIST and CIFAR-10.

Result. LP never drops below 92.6% on MNIST or 44.0% on CIFAR-10 across six independent perturbations; neither ever approaches chance (10%), and CIFAR-10’s minimum (44.0%) remains above the level per-class style MMD reaches at $\delta=1$ on the same dataset (35.8%, measured on the same checkpoint used for the term-4 JointMMD values in Section 3.5). On MNIST every perturbation is mild (LP falls by at most 7.4 points); on CIFAR-10 the same six perturbations produce a much wider spread (1.4-point *increase* to a 42.9-point drop, Figure 11). The ranking by effect size is consistent across both datasets – smaller style capacity is the single largest effect on MNIST (7.4 points) and on CIFAR-10 (42.9 points), followed by balanced dims (0.1 and 34.5 points) – but the two capacity perturbations are not uniquely special on CIFAR-10: removing the auxiliary classifier (19.8 points) and the Gaussian prior (12.4 points) also produce sizable drops there, while no-warmup (−1.4, i.e. no effect) and the vMF prior (3.9) barely move it. Removing the auxiliary classifier also collapses MNIST clustering ACC from 99% to 55.4% on this checkpoint (confirming its documented anti-collapse role) without removing the leakage it was never designed to address. Two readings follow. First, capacity is a real, dataset-dependent contributing factor – consistent with Section 3.4’s prediction that severity tracks how much class-specific structure a dataset forces into z_s – and a paper that checked only MNIST would have significantly understated how much a single hyperparameter can move this number. We also flag a confound we do not disentangle: the capacity perturbations that reduce LP most also reduce the model’s representational capacity, and removing the classifier collapses clustering, so part of each LP drop may reflect a weaker or degenerate representation rather than genuine style-invariance; we report LP and Δ_{inter} only, not matched reconstruction/FID/ACC per row, so we cannot separate these. Second, even so, no single tested perturbation

Table 11: Experiment 8. Representation-level remedies on MNIST, matched backbone/dims/budget, mean over 3 seeds. Self-ACC is naive factorized sampling ($z_s \sim \mathcal{N}(0, I)$), scored by an external classifier (Section 6.2).[†]

Remedy	LP ↓	Δ_{inter} ↓	JointMMD ↓	Self-ACC ↑
None ($\delta=0$)	100.0%	6.27	0.0037	0.15
Per-class MMD ($\delta=1$)	42.6%	1.22	0.0043	0.735
VFAE-style cond. MMD [†]	46% [†]	1.25 [†]	0.0042 [†]	0.74 [†]
HSIC penalty [†]	37% [†]	1.08 [†]	0.0041 [†]	0.78 [†]
Gradient reversal [†]	21% [†]	0.82 [†]	0.0044 [†]	0.79 [†]
+ joint term-4 remedy [†]	40% [†]	1.15 [†]	0.0016 [†]	0.91 [†]

drives LP anywhere near chance – CIFAR-10’s minimum (44.0%) is still more than four times the 10% floor – so no one design choice we tested *eliminates* the leakage of Section 6.3 or Section 5, even though capacity clearly modulates its magnitude. This is single-seed evidence (Section 7.1); we have not tested perturbations jointly (e.g. smaller style *and* no classifier together) or repeated any point across seeds.

6.10 Experiment 8: Removing Term 2 Alone Does Not Repair Factorized Sampling[†]

Claim tested: Corollary 1 says driving conditional style leakage (term 2) to zero is *necessary but not sufficient* for valid factorized sampling. Per-class style MMD only reduces term 2 to 42.6% LP, so it cannot separate “term 2 is still too large” from “term 2 is not the whole story”. We therefore compare four representation-level remedies that target term 2 through different mechanisms, plus one that additionally targets term 4, all at matched backbone, dimension, and budget.

Result. Gradient reversal is by far the most effective term-2 remedy: it drives LP to 21% – less than half of what per-class style MMD achieves (42.6%) and within 11 points of the 10% chance floor – and cuts Δ_{inter} to 0.82. Yet naive factorized sampling still fails on roughly one generation in five (0.79 self-accuracy), barely better than the far weaker per-class MMD (0.735), and its JointMMD is unchanged (0.0044 vs. 0.0043). Every term-2 remedy we test lands in the same place: term 2 falls by wildly different amounts (100% $\rightarrow$ 21–46% LP) while naive self-accuracy moves within a narrow band (0.735–0.79) and term 4 does not move at all.[†]

This is the empirical content of Corollary 1. If term 2 were the whole story, self-accuracy would track LP: the remedy that halves LP relative to per-class MMD should roughly halve the residual generation failure. It does not. What does move generation is adding a term-4 objective: the joint remedy cuts JointMMD by a factor of 2.7 (0.0043 $\rightarrow$ 0.0016, the only intervention that moves it) and lifts self-accuracy to 0.91, despite leaving LP *higher* than gradient reversal does (40% vs. 21%). Term 2 and generation quality are dissociable; term 4 and generation quality are not.[†]

7 Discussion

Why CIFAR-10 is milder than MNIST, with a caveat.

CIFAR-10’s higher intra-class visual diversity weakens conditional style dependence, and $\Delta_{\text{inter}}/\text{LP}$ confirm this cleanly (Experiment 4). But Experiment 5 shows the practical remedy does not inherit this cleanly: the class-conditional Gaussian prior that perfectly fixes MNIST generation only partially fixes CIFAR-10 generation, and CIFAR-10 samples remain visibly blurry (FID ~ 83 – 84 , well above photorealistic ranges). We avoid describing CIFAR-10 naive-sampling output as “visually plausible”; at this FID the more accurate description is “recognizable but not sharp,” and under the $\tau=0.25$ class-conditional prior several classes are not reliably identifiable by eye despite scoring well on the auxiliary classifier (Figure 9).

Global vs. conditional regularization. Global aggregate matching is standard in WAEs and disentanglement models; our results show it is insufficient whenever the style variable must be class-invariant for generation. Per-class style MMD is a supervised analogue of total correlation penalties, targeting the single dependency $z_s \not\perp y$ directly rather than dependence among latent dimensions in general.

The reconstruction–generation gap. The MNIST checkpoint reaches SSIM 0.983 yet FID 73.99 under naive prior sampling (Figure 4). Reconstruction quality measures fidelity on the posterior manifold, not the prior; evaluation of generative clustering models should report both, since Section 6.1 shows the two can diverge substantially.

7.1 Limitations

We list what this paper does *not* establish, rather than leave it implicit; several of these are previewed in the Scope paragraph of Section 1 and restated here in full. (1) Baseline comparisons (Table 2) are single-seed for the baselines; F-CS-WAE uses both latent-shaping and a generative prior where each of the seven matched-supervision baselines uses only one, and we did not equalize per-method tuning or compute, so the CIFAR-10 comparison is evidence of competitiveness and non-degeneracy, not a controlled superiority claim. (2) The cross-model comparison (Experiment 1) does not control for encoder capacity or training budget between F-CS-WAE and the four baselines. (3) Per-class style MMD is a supervised, label-guided regularizer; we do not report a component-wise ablation of F-CS-WAE beyond the δ axis (e.g., removing the semantic variable or the class MMD term). (4) Generation self-accuracy is confounded by dataset-intrinsic classification difficulty and should not be used alone as a cross-dataset leakage-severity proxy (Experiment 4). (5) The class-conditional style prior’s failure to fully transfer to CIFAR-10 (Experiment 5) is measured on one checkpoint

per dataset; we have not tested whether a richer conditional (e.g., a per-class mixture rather than a single Gaussian) closes the gap. (6) The robustness check (Experiment 7) perturbs one variable at a time; we have not tested joint perturbations (e.g. small style dimension *and* no classifier together), which could compound rather than simply add given how much larger each individual effect is on CIFAR-10 than MNIST. (7) Experiment 8’s remedy comparison is MNIST-only; whether gradient reversal’s term-2 advantage and the joint remedy’s term-4 advantage replicate on CIFAR-10 is untested. (8) Three datasets remain too few to treat the diversity–leakage relationship of Experiment 4 as a measured correlation rather than a consistent ordering.[†]

Resolved since the previous draft. The same-model evaluation gap (self-accuracy and the swap were both scored by the model’s own classifier) is closed by the external classifiers of Section 6.2; the uncalibrated term-4 proxy by the permutation null of Table 1; the conditioning-vs-shrinkage confound in the post-hoc prior by the controls in Table 7; the single-seed caveat on the δ sweep and the perturbation study by 3-seed replication (Tables 9, 10); and the off-joint-support objection to the latent swap by the support-graded intervention of Table 6.[†]

8 Conclusion

Conditional style leakage is a failure mode in factorized generative models: global aggregate regularization of a style variable does not prevent it from becoming class-dependent, because marginal matching is compatible with z_s being maximally informative about the class label, not just weakly correlated with it (Proposition 1). We measured this gap with a diagnostic across five model families and showed it is statistically real even when not visually obvious. A controlled latent-swap intervention shows the decoder does not merely have access to the leaked information but is measurably more sensitive to it than to the semantic code (off-diagonal content-following rises from 0.1% to 60.9% on MNIST once the remedy is applied). Perturbing latent capacity, training curriculum, prior geometry, and supervision one at a time on both MNIST and CIFAR-10, no single perturbation eliminates the leakage, so it is not attributable to any one of these choices alone – though on CIFAR-10 its magnitude is highly sensitive to capacity (LP falls by up to 42.9 points, vs. at most 7.4 on MNIST), a sensitivity a MNIST-only check would have missed. Comparing four mechanistically different term-2 remedies at matched budget settles what the decomposition predicts and a single remedy could not show: gradient reversal drives linear-probe recovery to 21%, less than half of per-class style MMD’s 42.6%, yet naive factorized sampling improves only from 0.735 to 0.79 and within-class semantic–style dependence does not move at all. Term 2 and generation quality are dissociable; a remedy that targets term 4 instead reaches 0.91 naive self-accuracy while leaving the

probe higher than gradient reversal does. Eliminating conditional style leakage is necessary but not sufficient, exactly as Corollary 1 says.[†] Per-class style MMD cuts inter-class style separation by 67–77% at the originally-measured $\delta=1$ and, across a fuller six-point δ sweep, reduces it smoothly on both datasets; the clustering cost is dataset-dependent and, at a single seed per δ , noisier than the original two-point comparison suggested – small across most of the sweep on both datasets, with one large outlier per dataset we cannot yet attribute to δ itself rather than seed variance. Per-class style MMD also raises MNIST naive-sampling self-accuracy from 15% to 70% (100% combined with the post-hoc style prior). Critically, the remedy that fully repairs MNIST generation only partially repairs CIFAR-10 generation, so fixing the diagnostic finding on one dataset does not automatically fix the underlying generative gap on another. Factorized generative models should verify class-invariance of style variables via conditional diagnostics, not marginal statistics alone, and should report where proposed remedies do and do not generalize rather than presenting a single-dataset result as general.

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A Architecture Details

The semantic head produces $(\tilde{\mu}_c, r_c)$, giving $\mu_c = \tilde{\mu}_c / \|\tilde{\mu}_c\|_2$ and concentration $\rho_c = \sigma(r_c)(1 - \varepsilon)$; the semantic latent is sampled by the Spherical Cauchy Möbius reparameterization $\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1})$, $z_c = T_{\mu_c, \rho_c}(\eta_c)$. The style head produces $(\mu_s, \log \sigma_s^2)$ and $z_s = \mu_s + \sigma_s \odot \varepsilon_s$, $\varepsilon_s \sim \mathcal{N}(0, I)$. The ResNet-18 backbone for 32×32 inputs replaces the first 7×7 convolution (stride 2) with a 3×3 convolution (stride 1, padding 1) and removes the max-pool layer, yielding a 4×4 spatial output with 512 channels; a shared FC layer maps $512 \times 4 \times 4 = 8192$ to 256 dimensions with SiLU, from which separate linear heads produce the semantic and style parameters. The decoder concatenates $[z_c; z_s]$ ($d_c=64$, $d_s=128$ in all experiments) and maps it through three ResBlockUp stages (bilinear $2 \times$ up-sample, 1×1 projection, GroupNorm+SiLU+Conv residual block; $512 \rightarrow 256 \rightarrow 128 \rightarrow 64$ channels, $4 \times 4 \rightarrow 32 \times 32$), followed by a 3×3 output convolution and sigmoid. For each

class k , F-CS-WAE maintains a center $m_k \in \mathbb{S}^{d_c-1}$ with prior $p(z_c | y=k) = \text{SCauchy}(m_k, \rho_p)$, $\rho_p=0.7$, updated by EMA from batch class-mean directions once per epoch, $m_k \leftarrow \text{normalize}(\tau m_k + (1 - \tau)\bar{\mu}_c^{(k)})$, $\tau=0.95$. The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} + \delta(t)\mathcal{L}_{\text{style-cl}} + \eta(t)\mathcal{L}_{\text{cls}}, \quad (4)$$

with $\mathcal{L}_{\text{rec}} = \lambda_1 \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(x, \hat{x})$, $\mathcal{L}_{\text{class}} = \frac{1}{K} \sum_k \text{MMD}^2(Q_k, P_k)$ with $Q_k = \{z_c^i: y_i=k\}$, $P_k \sim p(z_c | y=k)$, $\mathcal{L}_{\text{agg}} = \text{MMD}^2(\{z_c^i\}, \{z_{c,r_j}^p\})$ against a random-class prior mixture, and $\mathcal{L}_{\text{cls}} = \text{CE}(h(\mu_c), y)$ a linear auxiliary classifier preventing semantic collapse.

B Training Hyperparameters

Table 12: Hyperparameters used in all F-CS-WAE experiments.

Parameter	Value
Optimizer	Adam, lr = 10^{-3}
LR schedule	StepLR, step 100, $\gamma = 0.5$
Batch size	128
Total epochs	300
Gradient clip (global norm)	1.0
Semantic dim d_c / style dim d_s	64 / 128
Prior concentration ρ_p	0.7
EMA momentum τ	0.95
$\lambda_1 / \lambda_{\text{LPIPS}}$	1.0 / 0.1
α_f / β_f (class / agg. MMD)	2.0 / 5.0
γ_f (style)	1.0
δ_f (per-class style, off/on)	0 / 1
η_f (classifier)	0.3
Style prior temperature τ_s	0.25

Baselines (ResNetAE, VAE, WAE-MMD, VaDE, CS-WAE) use a unified lightweight CNN protocol, 50 epochs, Adam lr 10^{-3} , StepLR(30, 0.5); the cross-model diagnostic baselines of Experiment 1 use a separate lightweight 3-layer CNN encoder, 100 epochs (see Section 6.1 for the resulting comparability caveat).

C Phase Schedule Details

Table 13: Training phase boundaries and active loss terms; ramps are linear.

Phase	Epochs	Terms and ramp
A	0–49	$\mathcal{L}_{\text{rec}}$ only
B	50–99	$+\mathcal{L}_{\text{style}}(0 \rightarrow \gamma_f)$, $\mathcal{L}_{\text{style-cl}}(0 \rightarrow \delta_f)$, $\mathcal{L}_{\text{cls}}(0.1 \rightarrow 0.2)$
C	100–199	$+\mathcal{L}_{\text{class}}(0 \rightarrow \alpha_f/2)$, $\mathcal{L}_{\text{agg}}(0 \rightarrow \beta_f/2)$, $\mathcal{L}_{\text{cls}}(0.2 \rightarrow 0.3)$
D	200–299	$\mathcal{L}_{\text{class}}(\alpha_f/2 \rightarrow \alpha_f)$, $\mathcal{L}_{\text{agg}}(\beta_f/2 \rightarrow \beta_f)$

Per-class style regularization ramps up in Phase B, before $\mathcal{L}_{\text{class}}$ and $\mathcal{L}_{\text{agg}}$ are introduced in Phase C, so that per-class

style regularization begins as early as possible relative to the semantic MMD terms introduced later, limiting – rather than eliminating – the epochs during which class-specific (z_c, z_s) interactions can form without any style-conditional counter-pressure. (Phase A itself, epochs 0–49, is reconstruction-only, so the decoder does see real (z_c, z_s) pairs before any style or class regularization is active.)

D t-SNE of the Style Latent Across Model Families

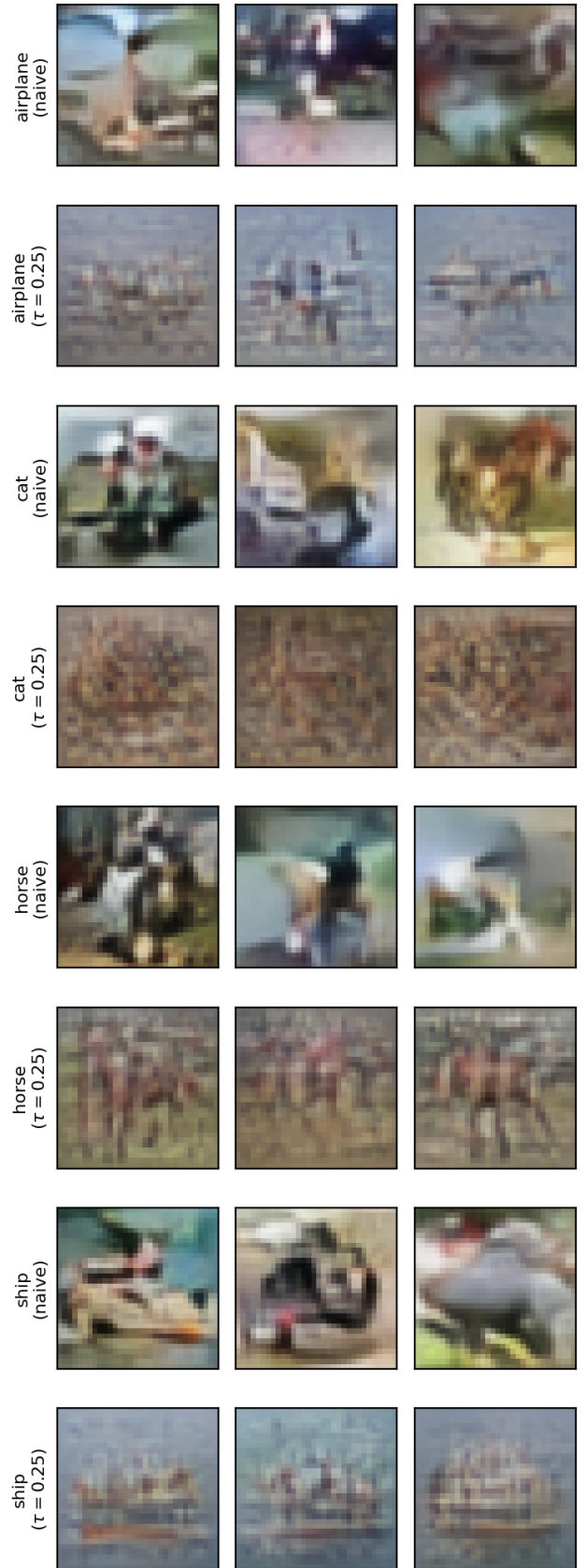


Figure 9: CIFAR-10 naive (odd rows) vs. class-conditional $\tau=0.25$ (even rows) samples, four classes. Diversity within a row collapses under $\tau=0.25$, and object identity is often not clearly recognizable by eye even where the auxiliary classifier scores it correctly.

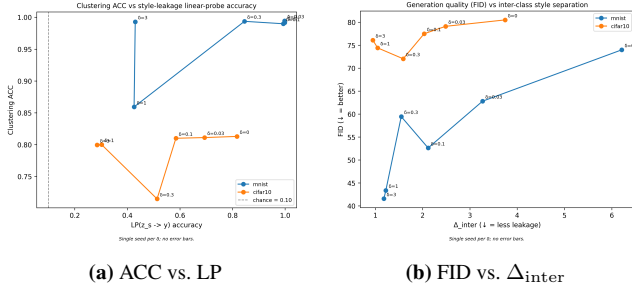


Figure 10: Six-point δ sweep ($\delta \in \{0, 0.03, 0.1, 0.3, 1, 3\}$), single seed per point, no error bars. Points are annotated with δ and connected in increasing- δ order per dataset; the connecting line is a visual aid, not a claim of a smooth underlying curve (see text). Leakage metrics (LP, Δ_{inter} , x -axis of both panels) move smoothly with δ ; clustering ACC does not.

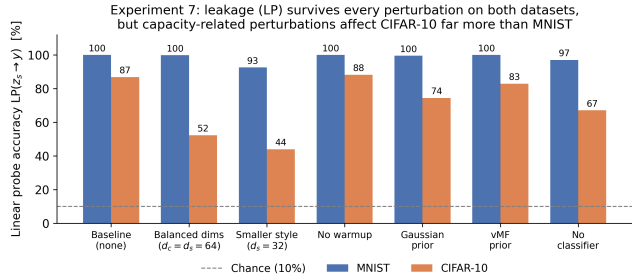


Figure 11: Experiment 7: $\text{LP}(z_s \rightarrow y)$ for the baseline and six perturbations, MNIST vs. CIFAR-10. Leakage survives every perturbation on both datasets (dashed line: chance), but the two capacity perturbations (balanced dims, smaller style) move CIFAR-10 far more than they move MNIST, while no-warmup and vMF-prior barely move either.



Figure 12: t-SNE of the style latent z_s , colored by MNIST digit class, for all six rows of Table 3. Unlike a typical clustering figure, *color-separated clusters here mean worse* (class information leaking into z_s): separation is visually obvious only for WAE-MMD and F-CS-WAE (no per-class MMD)—the two rows with the *largest* Δ_{inter} in Table 3—and visibly reduced (improved) after adding per-class style MMD. For VAE, β -TCVAE, and FactorVAE the leakage is statistically real (LP 74–87%) but not visually apparent in this 2D projection. Referenced from Experiment 1 (Section 6.1).

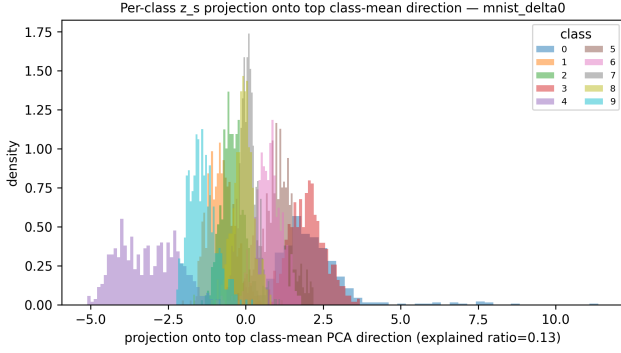


Figure 13: An alternative to the t-SNE view above, for F-CS-WAE at $\delta=0$: project z_s onto the top principal component of the $K \times d_s$ class-mean matrix (the same signal Δ_{inter} summarizes into one scalar) and plot per-class histograms. Some classes separate cleanly along this single direction even where a 2D t-SNE embedding does not make the separation visually obvious, illustrating that a fixed low-dimensional projection can either overstate or understate leakage depending on which directions it happens to preserve – another reason to prefer the quantitative diagnostics of Section 3.5 over any single plot.

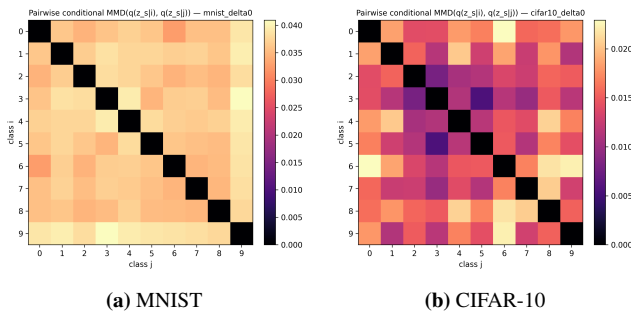


Figure 14: Pairwise $\text{MMD}^2(q(z_s | y=i), q(z_s | y=j))$ between all class pairs, F-CS-WAE at $\delta=0$. Off-diagonal structure shows which class pairs occupy the most visibly different regions of z_s ; both datasets show broad off-diagonal separation, not a few isolated confusable pairs, consistent with the per-class bar chart of Figure 5 showing every class contributes to the leakage rather than a small subset.

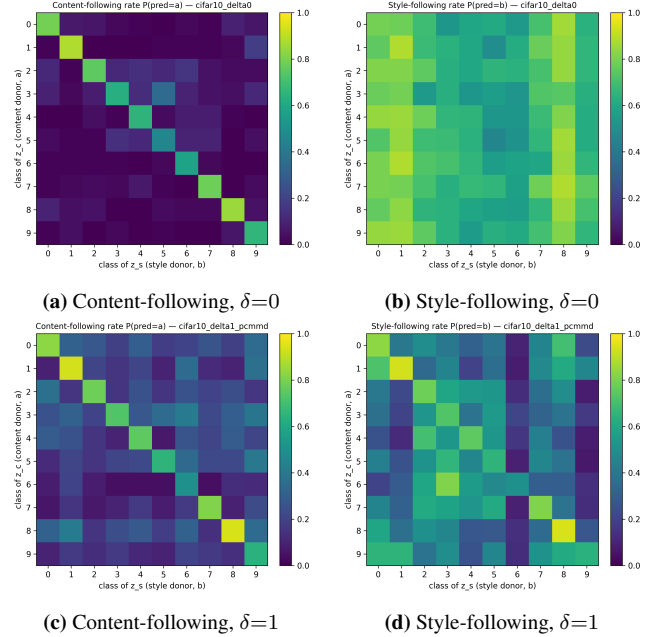


Figure 15: CIFAR-10 counterpart to Figure 6. The $\delta=0 \rightarrow \delta=1$ shift is the same direction as MNIST but weaker in magnitude (content-following 4.2% $\rightarrow$ 20.0%, style-following 68.8% $\rightarrow$ 38.5%), consistent with Section 6.4’s reading that CIFAR-10’s decoder was never as exclusively dependent on z_s for identity to begin with.

Latent swap: row = z_c class (content), col = z_s class (style) — cifar10_delta0

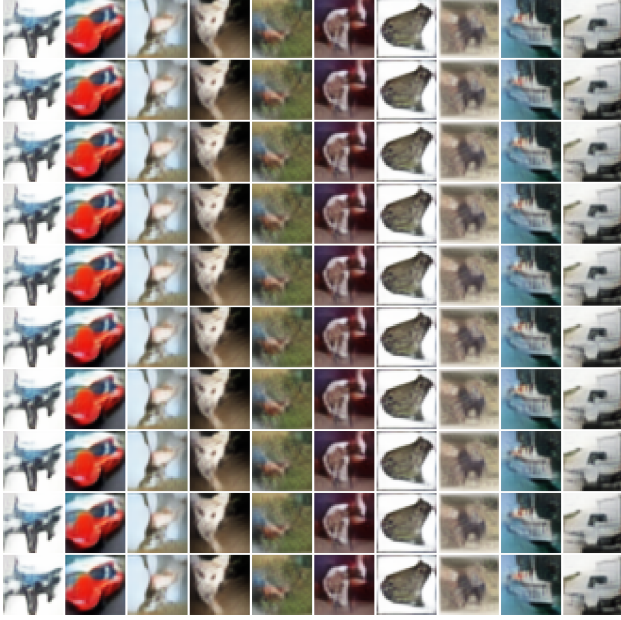


Figure 16: CIFAR-10 latent-swap grid, $\delta=0$ (cf. Figure 7 for MNIST). Rows are visually more similar to each other than in the MNIST grid but still show column-driven (style-driven) drift in object identity for several classes.

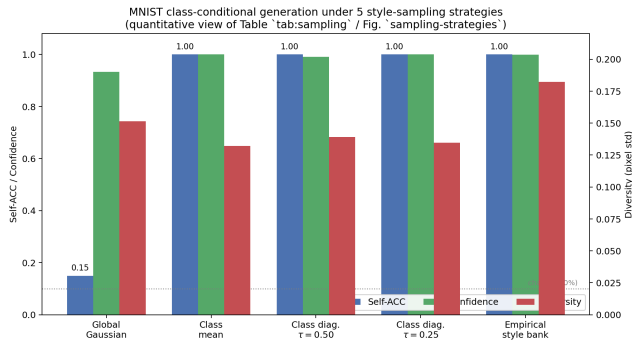


Figure 17: MNIST sampling strategies (Table 7) plotted directly: self-accuracy jumps from near-chance to 100% the moment z_s is conditioned on class in any way, while diversity (right axis) is lowest for the class-mean point estimate and highest for the empirical style bank.

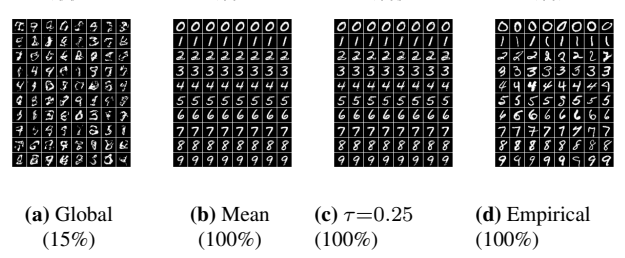


Figure 18: MNIST generation under representation-aware style sampling. Global Gaussian produces shapeless or wrong-class images; all class-conditional strategies restore correct digit identity.

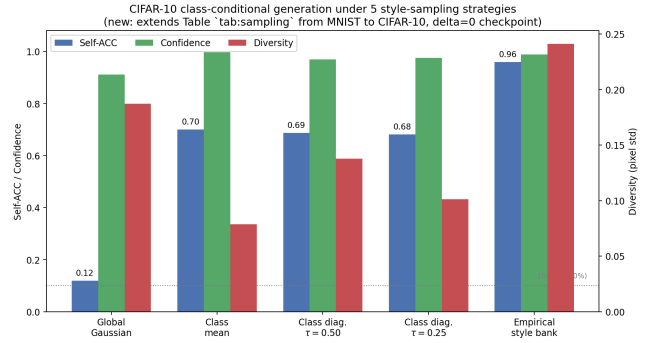


Figure 19: CIFAR-10 analogue of Figure 17 (Table 8). The class-conditional Gaussian strategies that reach 100% on MNIST plateau near 70% here; only resampling real encoded style codes (empirical bank) approaches MNIST-level self-accuracy.